College of Health Solutions
Arizona State University

Nutritional and Metabolic Resilience

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Comparative Physiology Lab
Nutritional & Metabolic Resilience

Unique Adaptations for Unique Solutions

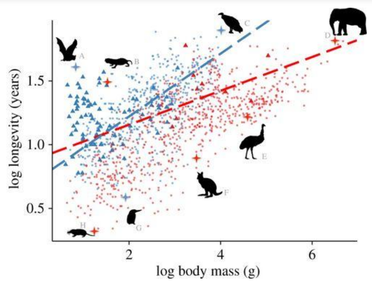
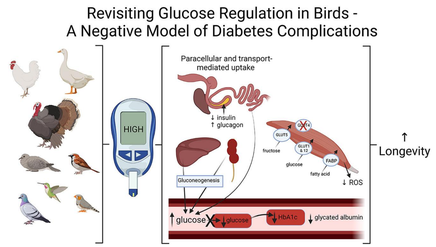


Figure from: Healy et al., 2014 (doi: 10.1098/rspb.2014.0298)

- Comparative physiology research utilizes animal models naturally suited to answer research questions.
- Birds live longer, healthier lives than mammals of similar body mass even though they have the highest blood sugar and lipid concentrations among vertebrates.
- We are exploring how birds protect themselves from developing diabetes and cancer to live extraordinarily long lives among vertebrates (Kapsetaki et al., PMID 37577544).
- We are also exploring the unique ways in which they regulate blood sugar and natural insulin resistance (Sweazea et al., 2022, PMID 35817273).

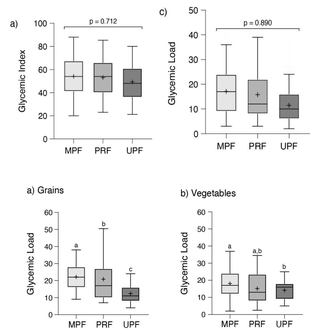


One Health: Food Quality, Availability, Access, and Choice in Urban Environments

Central Arizona-Phoenix Long-Term Ecological Research
CAP LTER

Figure from: Buffalo State College, NY

- The quality of available foods varies across the spectrum of urban environments and historical policies continue to affect food availability, quality, access, and choice for humans and animals.
- We analyzed 1,995 common food items and found that glycemic index and glycemic load do not differ across NOVA Classification food processing levels (Basile et al., 2024, PMID 39209154).
- However, our analysis of 1,946 food items support that minimally-processed foods have a higher total antioxidant content than processed or ultra-processed foods (Basile et al., 2024, PMID 39523204).
- We are also exploring food choice among wild birds across an urban-rural gradient and the impact of poor nutrition in the food chain (IACUC protocol 25-2115R).
- Similarly, we are conducting a pilot survey funded by the ASU Institute of Social Science Research to assess consumer perception on the safety and nutritional value of various sources of eggs that might drive purchasing behaviors.



Improving Diabetes and Health through Dietary Interventions

- Western diets high in saturated fats, simple carbohydrates, and salt are a major contributor to the development of metabolic syndrome and diabetes.
- My lab is also exploring a novel fulvic acid supplement to mitigate symptoms related to metabolic syndrome.